Yu-Ting Liao

computational artist & designer — the body as an interface

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ABOUT

Designer and computational artist exploring **the body as an interface** through wearable technology, physical computing, and embodied interaction. My practice brings together fashion pattern cutting, garment construction, hardware prototyping, machine learning, and creative coding. I recently completed an MA in Computational Arts at Goldsmiths, with previous degrees from the University of the Arts London and Shih Hsin University, and over three years of professional visual design experience in Taipei.

EDUCATION

Sep 2025 — Sep 2026

MA in Computational Arts, Distinction — Goldsmiths, University of London

Final project: **Solo Choir**.

Physical computing, embodied interaction, real-time audio and machine learning for creative practice (Raspberry Pi Pico, Arduino, Python, p5.js, Tone.js, TouchDesigner).

Sep 2022 — Jul 2024

MA in Digital Multimedia Arts — Shih Hsin University, Taipei

Graduate thesis: *Technology Inside AI Game Development and Application: An Example of Generative AI NPC*. Developed an AI NPC dialogue system in Unreal Engine 5.2 using GPT-4, integrating speech recognition, synthesis and sentiment analysis for context-aware, real-time NPC interaction.

Sep 2016 — Jun 2019

BA in Fashion Pattern Cutting — University of the Arts London

SELECTED PROJECTS

2026

Solo Choir — MA final project · real-time vocal instrument · Goldsmiths

Turns one singer into a choir. Sing a phrase and the system answers with harmony voices made by real-time neural voice conversion.

Designed and built a tabletop responsive instrument (four speakers, head model, boom microphone) and two wearable prototypes, an open-ear bone-conduction headband and an auxetic neck ring for feeling harmony in the body.

demo: vimeo.com/1223687812 · code: github.com/yvwHY/solo-choir · case study

2026

Fabri-Form — human–AI interface · pattern cutting · Goldsmiths

A translation tool between sketch and pattern. A multimodal LLM reads a hand-drawn sketch and fills a bounded parameter schema. The browser then generates a printable 1:1 pattern block. Six sliders expose every inferred parameter, so authorship stays with the designer. The current version outputs paper patterns.

demo: vimeo.com/1186248027 · live: yvwhy.github.io/fabriform · case study

2026

Gesture-Float — hardware · wireless control · Goldsmiths

A wireless flying unit controlled through arm movement. An arm-mounted Raspberry Pi Pico W measures orientation (pitch and roll) and transmits control data via Wi-Fi (UDP) to a helium balloon unit. The electronics were assembled using hand-soldered circuitry.

demo: vimeo.com/1176254975 · case study

2025

The Bonnet — wearable · computer vision · Goldsmiths

An anti-surveillance headpiece. A custom-trained YOLOv8 model detects a CCTV camera and an Arduino servo drops a fabric veil over the face.

demo: vimeo.com/1144358503 · case study

EXHIBITIONS

Aug 2026

Scream, Unsynced — MA Computational Arts Festival 2026, Goldsmiths, London

Postgraduate degree show, St James Hatcham Building, 27–29 Aug 2026. Exhibited Solo Choir.

May 2026

Bye Default — Computational Arts Pop-Up, PSH Atrium, Goldsmiths, London

Exhibited *Harmonizing!*, a browser-based live vocal-harmony instrument; designed the exhibition graphics and caption cards for all 37 works.

EXPERIENCE

Sep 2023 – Jun 2025

Visual Designer — Nice Crush Design House, Taipei

Created visual assets, including digital advertisements, social media graphics, packaging and promotional materials. Developed marketing strategies to ensure cohesive visual communication across both online and offline platforms.

Nov 2021 – Sep 2022

Visual Designer — Digibionic, Taipei

Designed and created visual assets for brand marketing.

Dec 2020 – Jun 2021

Graphic Designer — AY-AY International, Taipei

Created graphic assets and layouts for a wide mix of multimedia projects. Produced and maintained style guides.

SKILLS

making	pattern cutting, garment construction, circuit soldering
physical computing	Raspberry Pi Pico W, Arduino, servo/motor control, IMU orientation sensing, bone-conduction and surface-exciter audio
audio / machine learning	real-time neural voice conversion, DDSP, pitch detection, Web Audio API, YOLOv8, OpenCV, Python, custom dataset curation (Roboflow)
creative code	p5.js, Tone.js, ml5.js, TouchDesigner, JavaScript
3d / game engine	Unreal Engine 5, 3D computer graphics
design	Adobe Illustrator, Photoshop, InDesign, After Effects

references and full documentation available on request · yvwhy.github.io · updated Oct 2026